

GENAI ENGINEER ROADMAP

60 Days - Parts, Difficulty, Projects & Interview Milestones

Concise planning edition

Difficulty	Meaning	Mode	Meaning
1-4 Easy 5-6.5 Moderate 7-8 Hard 8.5-10 Very Hard	Estimated cognitive/engineering load	Concept / Applied / Project / Interview	Dominant focus for that day

Roadmap at a glance

Sprint	Days	Focus	Avg. difficulty	Major milestone
Sprint 0	1	Set up the engineering workflow.	2.0/10	Foundation complete
Sprint 1	2-16	Understand LLM internals from tokens to inference.	7.2/10	Project day(s): 16
Sprint 2	17-25	Build retrieval and production RAG capability.	7.3/10	Project day(s): 25
Sprint 3	26-34	Use LLM APIs, frameworks, agents and orchestration.	6.9/10	Project day(s): 34
Sprint 4	35-42	Build production backend services for GenAI systems.	6.8/10	Project day(s): 42
Sprint 5	43-50	Containerize, automate and deploy.	6.6/10	Project day(s): 50
Sprint 6	51-60	Build enterprise projects, debug production, prepare interviews.	8.4/10	Projects D51-60; interviews D59-60

Important: the roadmap source defines the topics. Difficulty scores and mode labels below are estimates added only for planning.

Topics from the user's roadmap. Difficulty and mode labels are planning estimates, not source ratings.

Sprint 0 + Sprint 1A - Engineering & LLM Foundations

Days 1-8 | Setup, tokenization, embeddings, attention math foundations

Day	Parts / topics	Mode	Difficulty
1	Repository Setup - Git Basics - GitHub - VS Code - Engineering Workflow - Documentation Structure - Learning Roadmap	Foundation	2/10
2	Introduction to LLMs - Traditional Programming vs AI - Tokenizer - Token IDs - Embeddings - Context - Attention (Concept)	Concept	5/10
3	Query - Key - Value - Query-Key Matching - Attention Scores - Softmax (Concept) - Weighted Values - Engineering Assessment	Concept	7/10
4	Why Embeddings aren't enough - Query from Embedding - Key from Embedding - Value from Embedding - Wq - Wk - Wv - Learned Transformations	Concept	7/10
5	Matrix Basics - Vectors - Matrix Multiplication - Vector Transformation - $Q = XWq$ - $K = XWk$ - $V = XWv$	Concept	7/10
6	Dot Product - Similarity - $Q \times K^T$ - Attention Score Calculation	Concept	7/10
7	Scaling ($\sqrt{dk}$) - Why Scaling is Needed - Numerical Stability	Concept	7.5/10
8	Softmax (Mathematics) - Attention Weights - Weighted Sum - Attention Equation	Concept	7/10

Average difficulty: 6.2/10.

Sprint 1B - LLM Foundations

Days 9-16 | Self-attention through inference, revision and mini project

Day	Parts / topics	Mode	Difficulty
9	Self-Attention - Attention Matrix - Information Flow	Concept	8/10
10	Multi-Head Attention - Multiple Heads - Concatenation - Output Projection	Concept	8/10
11	Positional Encoding - Sin/Cos Encoding - Position Information	Concept	6.5/10
12	Transformer Architecture - Encoder - Decoder - Residual Connections - Layer Normalization	Concept	8/10
13	GPT Architecture - Decoder-only Transformer - Masked Attention - Next Token Prediction	Concept	8/10
14	Pretraining - Fine-Tuning - Transfer Learning - RLHF	Concept	7.5/10
15	Inference - Context Window - KV Cache - Greedy Search - Beam Search - Temperature - Top-K - Top-P	Concept	7/10
16	Sprint 1 Revision - LLM Interview Questions - Mock Interview - LLM Mini Project	Review + Mini Project	7/10

Average difficulty: 7.5/10. Project-focused: Day 16.

Sprint 2 - RAG & Retrieval Engineering

Days 17-25 | Embeddings, vector databases, retrieval, advanced/production RAG

Day	Parts / topics	Mode	Difficulty
17	Embedding Models - Embedding APIs - Similarity Search	Applied	6/10
18	Vector Databases - FAISS - ChromaDB - Pinecone	Applied	7/10
19	Chunking - Metadata - Indexing	Applied	6.5/10
20	Semantic Search - Hybrid Search - BM25	Applied	7.5/10
21	Retrieval - Top-K Retrieval - Context Building	Applied	6.5/10
22	RAG Architecture - End-to-End Flow	Applied	7/10
23	Advanced RAG - Query Expansion - Compression - Reranking	Applied	8/10
24	Production RAG - Evaluation - Hallucination - Guardrails	Production Applied	8.5/10
25	Sprint 2 Project - Enterprise RAG Chatbot	Project	8.5/10

Average difficulty: 7.3/10. Project-focused: Day 25.

Sprint 3 - LLM Frameworks

Days 26-34 | Prompting, model APIs, LangChain, LangGraph, MCP and agents

Day	Parts / topics	Mode	Difficulty
26	Prompt Engineering - Prompt Patterns	Applied	5.5/10
27	OpenAI API - Gemini API - Claude API	Applied	5.5/10
28	LangChain Basics	Framework	6/10
29	LangChain Chains - Memory - Tools	Framework	6.5/10
30	LangGraph Basics	Framework	7/10
31	MCP (Model Context Protocol)	Framework	7.5/10
32	AI Agents - Agent Loop	Framework	7.5/10
33	Multi-Agent Systems	Framework	8/10
34	Sprint 3 Project - Multi-Agent Assistant	Project	8.5/10

Average difficulty: 6.9/10. Project-focused: Day 34.

Sprint 4 - Backend Engineering

Days 35-42 | FastAPI, databases, auth, Redis, monitoring and production backend

Day	Parts / topics	Mode	Difficulty
35	FastAPI Basics	Backend	5.5/10
36	REST APIs - Dependency Injection	Backend	6/10
37	PostgreSQL	Backend	6.5/10
38	SQLAlchemy - Alembic	Backend	7/10
39	Authentication - JWT - OAuth	Backend	7.5/10
40	Redis - Background Tasks	Backend	7/10
41	Logging - Monitoring - Error Handling	Backend	7/10
42	Sprint 4 Project - Production Backend	Project	8/10

Average difficulty: 6.8/10. Project-focused: Day 42.

Sprint 5 - Production Engineering

Days 43-50 | Containers, Git workflow, CI/CD, Kubernetes, cloud and deployment

Day	Parts / topics	Mode	Difficulty
43	Docker	Production	5.5/10
44	Docker Compose	Production	6/10
45	Git Workflow - Branching - Pull Requests - Code Reviews	Production	6/10
46	CI/CD	Production	7/10
47	Kubernetes Basics	Production	7.5/10
48	AWS Basics	Production	6.5/10
49	Azure Basics	Production	6.5/10
50	Deployment Project	Project	8/10

Average difficulty: 6.6/10. Project-focused: Day 50.

Sprint 6 - Enterprise Projects + Career Readiness

Days 51-60 | Enterprise builds, monitoring, production debugging, interviews and capstone

Day	Parts / topics	Mode	Difficulty
51	Enterprise RAG Chatbot	Enterprise Project	8.5/10
52	AI Resume Screening System	Enterprise Project	8/10
53	AI Customer Support Agent	Enterprise Project	8.5/10
54	AI Meeting Assistant	Enterprise Project	8/10
55	AI Research Agent	Enterprise Project	8/10
56	Multi-Agent Enterprise Platform	Enterprise Project	9/10
57	Production Monitoring	Production Ops	8/10
58	Debugging Production Issues	Production Ops	9/10
59	System Design Interview - GenAI Interview	Interview	8.5/10
60	Final Capstone Project - Resume Review - Portfolio Review - Mock Interview - Job Application Strategy	Capstone + Career	9/10

Average difficulty: 8.4/10. Project-focused: Days 51, 52, 53, 54, 55, 56, 60. Interview/career-focused: Days 59, 60.

Milestones & Difficulty Curve

Milestone	Roadmap day(s)	Outcome	Typical difficulty
LLM foundation checkpoint	16	Revision + LLM mini project + mock interview	7/10
Enterprise RAG project	25	End-to-end retrieval system	8.5/10
Multi-agent project	34	Agent orchestration project	8.5/10
Production backend project	42	Backend system ready for AI workloads	8/10
Deployment project	50	Containerized/cloud deployment workflow	8/10
Enterprise project stretch	51-56	Six portfolio-grade AI systems	8-9/10
Production readiness	57-58	Monitoring + debugging real failure modes	8-9/10
Interview + capstone	59-60	System design, GenAI interview, final capstone, portfolio/job prep	8.5-9/10

How the difficulty changes

- Days 1-16: concept-heavy. The burden is understanding internals and first principles.
- Days 17-34: implementation-heavy. New concepts continue, but more learning happens by building RAG and agent systems.
- Days 35-50: engineering-heavy. Difficulty shifts to APIs, databases, reliability, deployment and integration.
- Days 51-60: project/interview-heavy. The hardest work is design, debugging, production judgment and explaining decisions.

Guiding principle: do not learn technologies in isolation - learn to solve engineering problems.